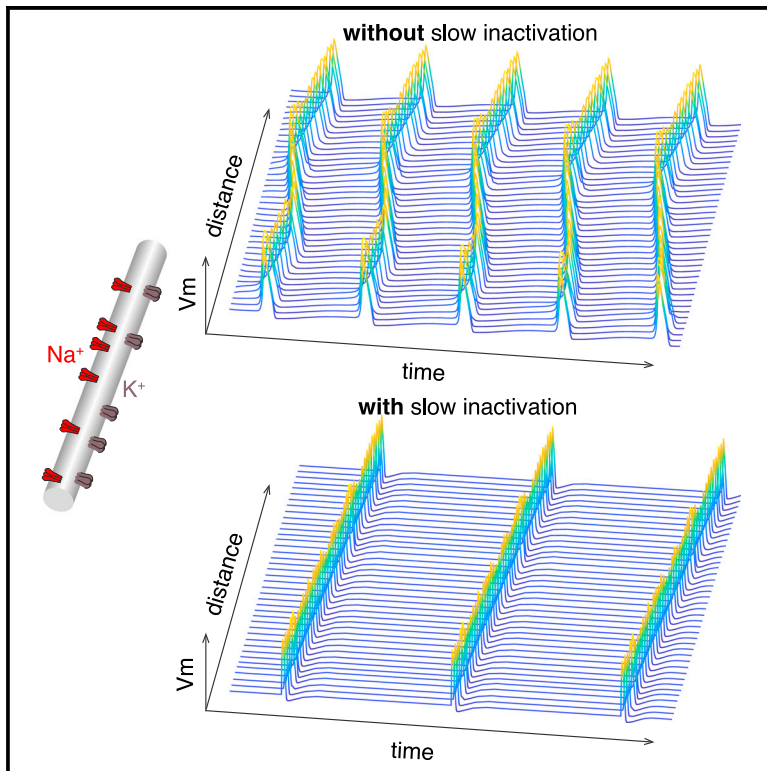


Current Biology

Sodium channel slow inactivation normalizes firing in axons with uneven conductance distributions

Graphical abstract



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In brief

Zang et al. model spike propagation in axons with uneven distributions of Na^+ and K^+ channels. Including slow inactivation of the Na^+ channel enhances the reliability of spike propagation and can compensate for uneven channel distributions.

Highlights

- Variation in ion channel density in axons may compromise axonal spike propagation
- Na^+ channel slow inactivation increases the reliability of spike propagation
- Normalization by slow inactivation can compensate for uneven channel distributions
- The normalization effect depends on inactivation kinetics and firing frequency

Report

Sodium channel slow inactivation normalizes firing in axons with uneven conductance distributions

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SUMMARY

The Na⁺ channels that are important for action potentials show rapid inactivation, a state in which they do not conduct, although the membrane potential remains depolarized.^{1,2} Rapid inactivation is a determinant of millisecond-scale phenomena, such as spike shape and refractory period. Na⁺ channels also inactivate orders of magnitude more slowly, and this slow inactivation has impacts on excitability over much longer time-scales than those of a single spike or a single inter-spike interval.^{3–10} Here, we focus on the contribution of slow inactivation to the resilience of axonal excitability^{11,12} when ion channels are unevenly distributed along the axon. We study models in which the voltage-gated Na⁺ and K⁺ channels are unevenly distributed along axons with different variances, capturing the heterogeneity that biological axons display.^{13,14} In the absence of slow inactivation, many conductance distributions result in spontaneous tonic activity. Faithful axonal propagation is achieved with the introduction of Na⁺ channel slow inactivation. This “normalization” effect depends on relations between the kinetics of slow inactivation and the firing frequency. Consequently, neurons with characteristically different firing frequencies will need to implement different sets of channel properties to achieve resilience. The results of this study demonstrate the importance of the intrinsic biophysical properties of ion channels in normalizing axonal function.

RESULTS AND DISCUSSION

Ion channels have a complex and rich set of slow processes that can capture them in inactivated states, removing them from the easily activated pool for as long as minutes or hours.^{15–24} We explore the hypothesis that slow inactivation of axonal Na⁺ channels may be a normalizing factor, contributing to resilience of spike propagation in axons^{11,12,25} with heterogeneous channel densities.

Slow inactivation

Slow inactivation reflects an activity-dependent transition of an ion channel between two pools of states: a pool that is available for activation by membrane voltage within the millisecond time-scale of spike generation (depicted *A* in Figure 1A), and a pool that is not available for activation, depicted (*1-A*). The states of a channel in the unavailable pool are coupled by transition rates operating in timescales extending from tens of milliseconds to many minutes. Recent studies demonstrate that at the protein population level, the unavailable (i.e., slowly inactivated) pool is a dynamic reservoir that regulates membrane excitability via control of the effective number of ion channels that participate in action potential generation.^{9,26}

Parameter variability across cells and within a cell

There is a growing appreciation that neurons of the same cell type can be degenerate, that is, produce similar activity with

variable sets of conductance densities.^{25,27–29} These degenerate solutions can arise naturally from the self-organizing homeostatic processes that govern the development and maintenance of the intrinsic electrical properties of neurons.^{30–32} Less attention has been paid to understanding the variance of ion channel densities along axons. Ion channel densities are not evenly distributed down the length of axons, and changes in axonal distribution of ion channels undoubtedly occur in disease.¹⁴ The purpose of this study is to understand the tolerance of action potential propagation to uneven distributions of ion channels and the role of Na⁺ channel slow inactivation in compensating for variability in channel distributions.

Na⁺ channel slow inactivation in a single compartment model

The impact of slow inactivation on a single compartment canonical Hodgkin-Huxley (H-H) model¹ is demonstrated in Figure 1. The slow *A* ↔ (*1-A*) kinetics is modeled with forward (γ) and backward (δ) rate constants that are scaled versions of the rates governing the standard rapid inactivation in the H-H model. Thus, $\delta(V) = 0.1 \times \alpha_h(V)$ and $\gamma(V) = 0.1 \times \beta_h(V)$ (see STAR Methods). The resulting time constant of the *A* ↔ (*1-A*) transition as a function of membrane potential is shown in Figure 1A (right). Figure 1B shows examples of the effects of slow inactivation on spiking in this toy model. As expected, at high maximal sodium conductance, the model produces tonic spiking, which may be

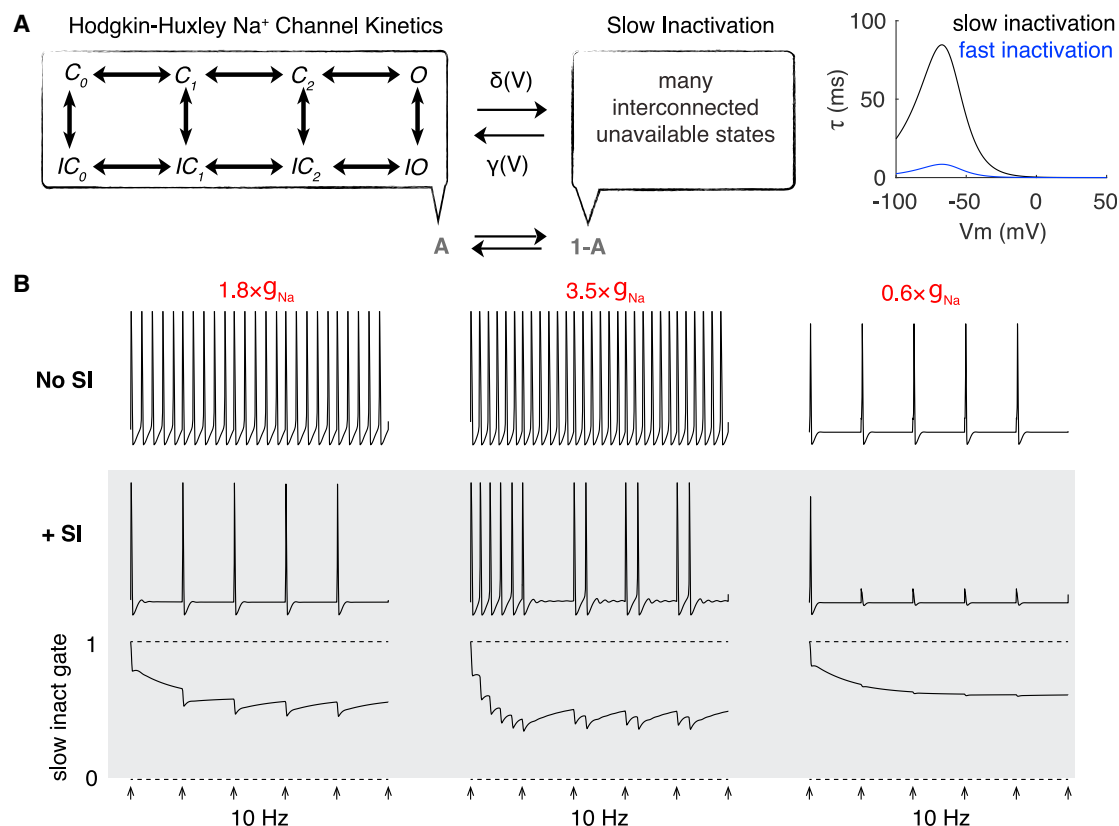


Figure 1. The effect of the Na⁺ channel slow inactivation gate on axonal excitability in a single compartment model

(A) A schematic of Na⁺ channel slow inactivation gate (left) and its time constant (right, black). The blue trace shows the time constant of the h gate in the H-H model.

(B) Arrows on the x axis indicate stimulation times (10 Hz). The top panel has no slow inactivation (no SI). Slow inactivation (+SI) was included in the shaded panels. At $1.8 \times g_{Na}$ (left) with no SI, the axon fires more rapidly than the stimulation frequency, but with SI, the axon faithfully follows the stimuli (normalization). At $3.5 \times g_{Na}$ (middle) the axon again fires more rapidly than the stimuli, but the addition of SI produces burst-like interruptions of activity. At $0.6 \times g_{Na}$ (right), the axon faithfully follows the stimulation frequency in the absence of SI, but with SI, it fails to follow repeated stimuli.

normalized by the introduction of slow inactivation (Figure 1B, left). The effect of the slow inactivation gate depends on the extent of deviation from the standard H-H maximal sodium conductance (see, for example, middle and right panels of Figure 1B). Note that while the H-H standard parameter for the sodium conductance is 120 mS/cm^2 , their measured range was $65\text{--}260 \text{ mS/cm}^2$.

Spike propagation in multicompartmental axonal models with even and uneven distributions of conductances

We built axonal multicompartment models that differed in their Na⁺ and K⁺ channel densities, as illustrated in Figure 2A (top). In any given compartment, the Na⁺ (red) and K⁺ (dark brown) conductance densities could be either above or below the values in the H-H model. In these studies, we stimulated the axon at one end at a fixed frequency and determined the extent to which the entire axon followed the stimulus. Many sets of parameters generated “faithful propagation” (Figure 2A, bottom left), in that each stimulus generated a propagating action potential. Other parameters resulted in “propagation failure” (Figure 2A, bottom center). Still other sets of parameters

produced “tonic activity,” when the axon was active either spontaneously or between stimuli. This occurs when the maximal sodium conductance in one or more compartments is high enough to cause tonic spiking. In the simulation shown in Figure 2A (bottom right), the axon was stimulated at 1 Hz at one end, but was tonically active at $\sim 50 \text{ Hz}$, in this case from an ectopic site (high Na⁺ conductance around this location) of action potential generation that triggered spikes that then propagated in both directions.

To directly compare the effects of uneven and even channel distributions, we randomly sampled scaling factors for the Na⁺ and K⁺ channel conductance densities (baseline is the H-H model) from Gaussian probability density functions with different variances (0.04, 0.09, 0.16, 0.25). For each variance, we simulated a large population of axons (Figure 2B). The mean of the Gaussian function was 1 (i.e., the H-H value) for all groups (Figure 2B, right). The histograms in Figure 2B show that at low variances, most axon models showed faithful propagation, but at the lowest variance, the uneven case was more reliable than the even case, presumably because there were axons in which neighboring compartments were able to compensate for compartments with Na⁺ conductances that were insufficient to

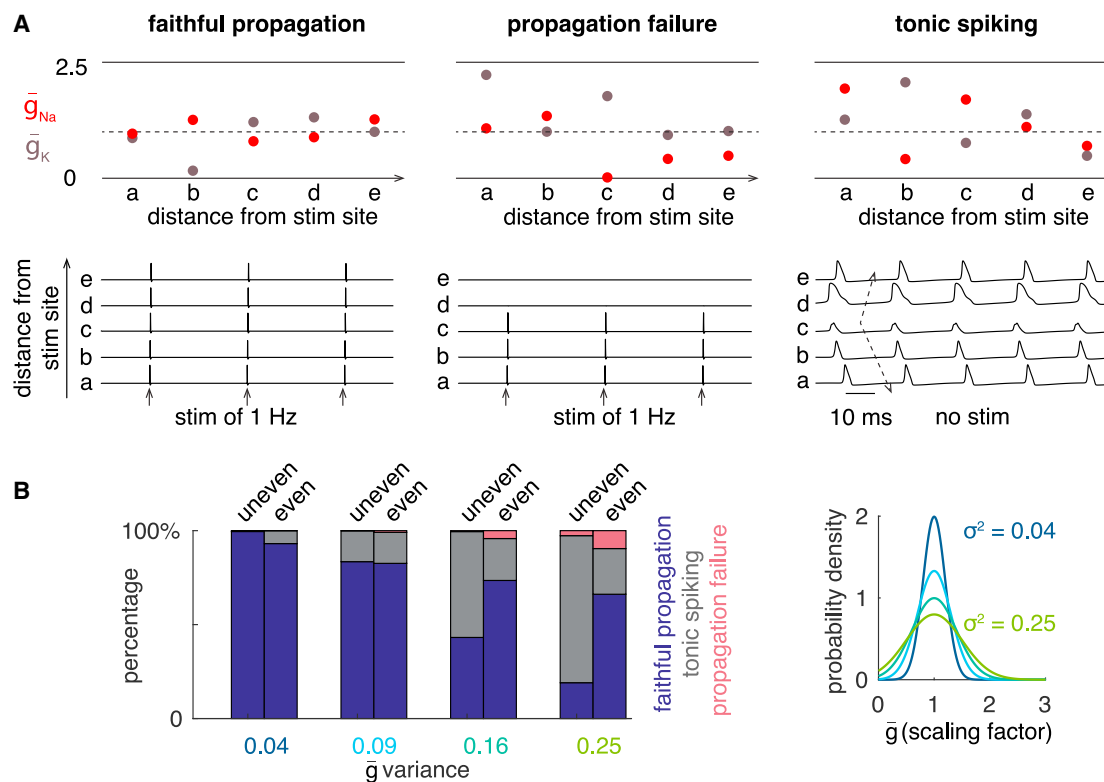


Figure 2. Abnormal spike generation and propagation in axons with varied ratios of channel conductance densities

(A) Examples of faithful propagation (left), propagation failure (middle), and tonic spiking (right) in axon models with uneven channel conductance densities. In the left two plots, models were stimulated at 1 Hz, but there is no stimulus during the time-range shown in the right plot. The above panels show the relative distribution of Na^+ (red) and K^+ (dark brown) conductance density scaling factors at corresponding sites in the bottom panel (0.5, 1.5, 2.5, 3.5, and 4.5 mm distant from axonal starting end). The dashed line represents a uniform H-H model ($\bar{g}_{\text{Na}} = 1$; $\bar{g}_{\text{K}} = 1$).

(B) The proportion of model behaviors with increased variance of channel conductance densities. For each variance value, the left (right) bar corresponds to axon models with uneven (even) distributed conductance densities. Na^+ and K^+ conductance density scaling factors were varied by randomly sampling scaling factors from Gaussian distribution functions with different variances (right).

maintain faithful propagation. As the conductance density variations increased, the number of cases that showed faithful propagation decreased and the number of cases that showed tonic spiking dramatically increased, especially in axons with uneven distributions of conductances. At higher variances, cases of propagation failure appeared, most notably in the axons with even distributions of conductances.

Slow Na^+ channel inactivation in multicompartmental models of axons with even and uneven conductance distributions

Next, we explored whether incorporating a slow inactivation gate could normalize spike generation and propagation by reducing axonal over-excitability. Because slow inactivation is intrinsically complex, we adopted the simplified formulations of Migliore³³ (see STAR Methods) to systematically manipulate the inactivation properties and study their effects on the normalization of axonal propagation. Using this model, the steady-state inactivation level, and the time constants of inactivation and recovery from inactivation, can be varied relatively independently (Figure 3A).

We used the same set of models as in Figure 2 to test the impact of slow inactivation on axonal spiking. Slow inactivation

obviously cannot increase spiking reliability for those neurons showing faithful propagation or propagation failure (examples not shown). Figure 3B shows three types of responses following the introduction of slow inactivation to axon models that produce tonic spiking. Initially, the gate was clamped to 1 in all three examples (slow inactivation was turned off), and they all produced tonic spiking. Once the slow inactivation gate is allowed to change dynamically with spikes, the axonal excitability consistently decreases. If the reduction is in the functional range, tonic spiking will not occur, and axons faithfully spike and propagate only when stimulated (Figure 3B, top). However, if the reduction is insufficient or excessive, the axons may produce either irregular firing (Figure 3B, middle, qualitatively similar phenomena have been observed experimentally³) or propagation failure (Figure 3B, bottom). Statistics include neurons showing faithful propagation and propagation failure before incorporating the slow inactivation gate.

As shown in Figure 3C, incorporating the slow inactivation gate makes most axon models faithfully spike in response to stimuli. When compared with Figure 2C, incorporating the slow inactivation gate did not substantially improve the propagation reliability of axon models with even conductance density distributions.

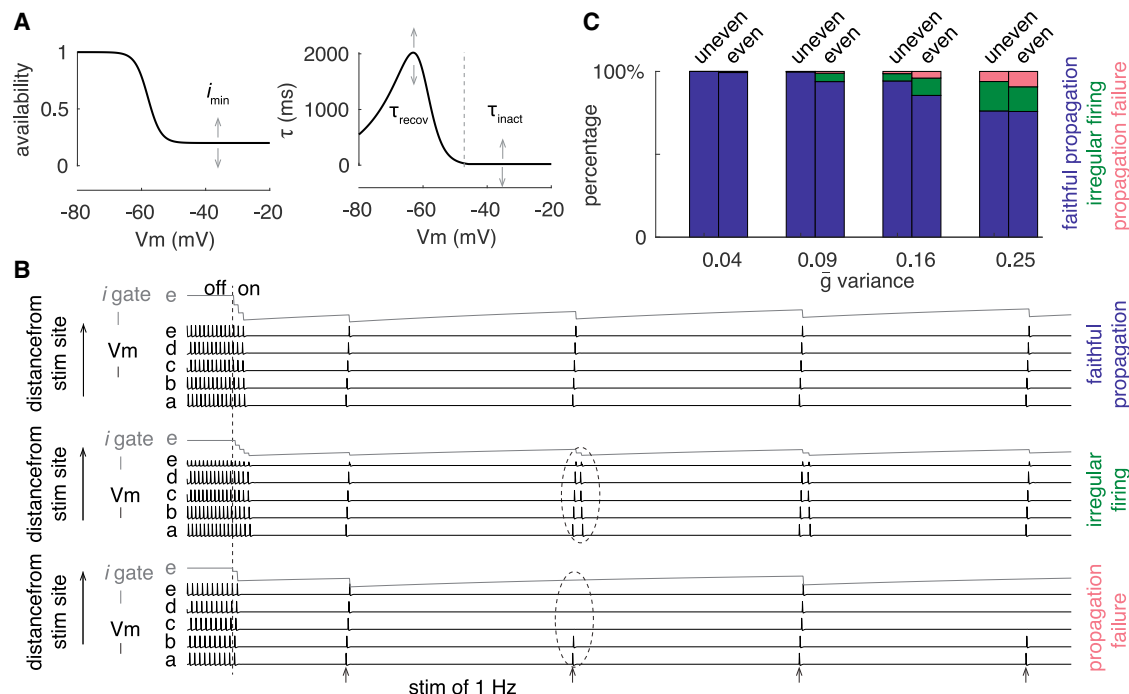


Figure 3. The normalizing effect of the Na^+ channel slow inactivation gate

(A) The steady-state inactivation curve (left) and time constants (right) of the i gate. Arrows show parameters whose variations are explored in Figure 4. (B) Typical examples of spike propagation pattern changes before and after switching on the i gate. In each plot, black traces represent V_m s at different distances from the stimulation site (as in Figure 2). Gray trace shows the i gate at the distally recorded axonal site. Scale bar of i gate is 0.6, and scale bar of V_m is 100 mV. (C) The proportion of propagation patterns in axon models with varying degrees of conductance density variances, after including the slow inactivation gate. For each variance value, the left (right) bar corresponds to axon models with uneven (even) distributed conductance densities. Color codes the propagation patterns.

The normalization of spiking by Na^+ channel slow inactivation is frequency-dependent

We explored how the three parameters (Figure 3A) of the slow inactivation gate affect its capacity to normalize axonal spike propagation. These parameters represent the steady-state availability of the slow inactivation gate (i_{min}), the inactivation time constant (τ_{inact} , time constants at depolarized membrane potentials), and the recovery time constant (τ_{reco} , time constants at hyperpolarized membrane potentials).

Each model was stimulated at 1 and 10 Hz. The normalizing effect of the slow inactivation gate depends on the stimulation frequency, as seen in the example traces in Figure 4A. This model fired tonically in the absence of slow inactivation. When the slow inactivation was turned on and the axon was stimulated at 1 Hz, the model faithfully followed. However, when the same model was stimulated at 10 Hz, propagation continued for about the first second and then failed. This occurred as a consequence of the slow inactivation that accumulated because of the 10 Hz firing. Interestingly, after about 2 s of silence a single spike was propagated as the inactivation started to wane.

We randomly generated models that produced tonic spiking without a slow inactivation gate and then systematically explored how inactivation gate parameters affect their normalizing effect on these over-excitable axons. Figure 4B presents a comparison of the effects of frequency on a population of axons with high variance. The first panel shows that as τ_{inact} increases, the rate of faithful spike propagation decreases for stimuli at 1 Hz but increases for 10 Hz stimulation. The 10 Hz

stimulation results in variable amounts of propagation failure, while the 1 Hz stimuli result in a relatively high proportion of irregular firing. Similar results are seen when i_{min} is increased. In contrast, when the τ_{reco} scaling factor increases, faithful propagation increases at 1 Hz but decreases at 10 Hz.

The frequency-dependent normalizing effects essentially reflect a non-trivial trade-off between the kinetics of slow inactivation and the excitability status of the axon. This trade-off is activity-dependent over a wide range of timescales, as the time constant of recovery from the unavailable (slow-inactivated) pool is related to the duration of previous activation by a power law.^{7–9,21,23}

Concluding remarks

Different neuronal types show characteristic firing rates, with some that usually fire at very slow rates and others much more quickly. They also have different spike shapes and constellations of ion channels.^{34–36} Essentially, slow inactivation reduces membrane over-excitability. In this work, we tested the impacts of slow inactivation on axons with different degrees of uneven channel distribution¹⁴ in the H-H model.¹ Figure 4 suggests that the role of slow inactivation in normalization of firing rates must match the specifics of the action potential mechanisms and ion channels in each type of neuron.^{37–39} Thus, cell-type definition should include the specification of the channel properties that can optimally implement normalization by slow inactivation. If a similar study of the role of slow Na^+ channel inactivation were to be implemented in axonal models with fast transient,

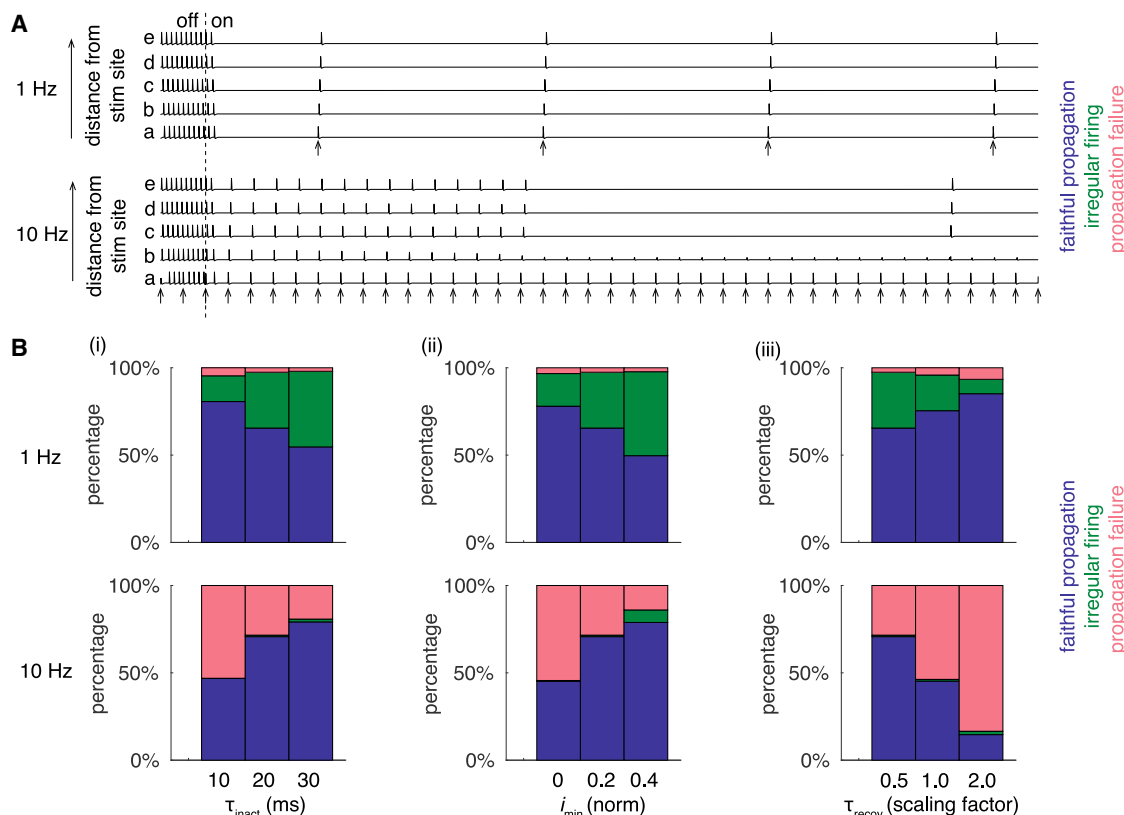


Figure 4. Frequency-dependent normalizing effect of the slow inactivation gate

(A) The effect of the slow inactivation gate on the propagation of spikes at different frequencies. Top to bottom, 1 and 10 Hz, respectively. Traces represent V_ms at different sites with distance from the stimulation site (as in Figure 2). Before the dashed vertical line, the *i* gate was clamped to be 1.

(B) The complex relationships among the *i* gate parameters, firing frequency, and the normalizing effect. The top and bottom row correspond to stimulation signals at 1 and 10 Hz, respectively. (i) τ_{inact} was varied, while $i_{min} = 0.2$ (norm), $\tau_{recov} = 0.5$ (scaled); (ii) i_{min} (norm) was varied, while $\tau_{inact} = 20$ ms, $\tau_{recov} = 0.5$ (scaled); (iii) τ_{recov} (scaled) was varied, while $\tau_{inact} = 20$ ms, $i_{min} = 0.2$ (norm). The axon models have a conductance density variance of 0.25. Color codes propagation patterns.

A-type K⁺ currents, or other types of voltage-gated channels that are sometimes found in axons,¹³ we would predict that the effects of Na⁺ channel inactivation on propagation would be qualitatively similar, but quantitatively different. Despite data indicating that ion channels are unevenly distributed,¹⁴ for most neuronal types, detailed descriptions of the spatial distributions of ion channel densities are lacking.

Axonal over-excitability will be influenced by channel distributions as well as axonal diameter.⁴⁰ In thin, unmyelinated axon models, spontaneous spiking elicited by uneven channel distributions occurs frequently and slow inactivation can efficiently normalize the firing (data not shown). It should be noted that this normalizing effect is not only limited to uneven distribution caused by local over-excitability, as explored here, but will also apply to neuronal over-excitability because of varied ratios of sodium/potassium channels at the cellular level.^{25,27–29} Here, we focused on the effects of slow Na⁺ channel inactivation on the reliability of spike propagation, but it may also function as a leaky integrator to constrain neurons at lower firing rates.⁴¹

Neuronal resilience can be enhanced in numerous ways, including the specifics of neuronal morphology.^{29,35} Here, we show another example of enhanced neuronal resilience that depends on the features of ion channel proteins but does not require

idiosyncratic tuning. Because we looked at populations of neurons with similar behaviors, the principle illustrated here holds across degenerate, or multiple, solutions. The effects of variable channel densities across extended morphologies can be complex,¹⁴ and slow channel inactivation is likely to be an important mechanism that is employed by neurons to enhance normalization and thus resilience. These results demonstrate that robust neuronal firing activities may be a direct result of the properties of proteins and not demand regulation mechanisms that involve signal transduction, transcription, and translation.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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AUTHOR CONTRIBUTIONS

Conceptualization, Y.Z., E.M., and S.M.; model simulation, Y.Z.; analysis and visualization, Y.Z.; writing – original draft, Y.Z., E.M., and S.M.; writing – review & editing, Y.Z., E.M., and S.M.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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REFERENCES

- Hodgkin, A.L., and Huxley, A.F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiol.* 117, 500–544. <https://doi.org/10.1113/jphysiol.1952.sp004764>.
- Hille, B. (2001). *Ion Channels of Excitable Membranes*, Third Edition (Sinauer).
- Fleiderer, I.A., Friedman, A., and Gutnick, M.J. (1996). Slow inactivation of Na^+ current and slow cumulative spike adaptation in mouse and guinea-pig neocortical neurones in slices. *J. Physiol.* 493, 83–97. <https://doi.org/10.1113/jphysiol.1996.sp021366>.
- Vilin, Y.Y., and Ruben, P.C. (2001). Slow inactivation in voltage-gated sodium channels: molecular substrates and contributions to channelopathies. *Cell Biochem. Biophys.* 35, 171–190. <https://doi.org/10.1385/CBB:35:2:171>.
- Blair, N.T., and Bean, B.P. (2003). Role of tetrodotoxin-resistant Na^+ current slow inactivation in adaptation of action potential firing in small-diameter dorsal root ganglion neurons. *J. Neurosci.* 23, 10338–10350. <https://doi.org/10.1523/JNEUROSCI.23-32-10338.2003>.
- Ulbricht, W. (2005). Sodium channel inactivation: molecular determinants and modulation. *Physiol. Rev.* 85, 1271–1301. <https://doi.org/10.1152/physrev.00024.2004>.
- Silva, J.R., and Goldstein, S.A. (2013). Voltage-sensor movements describe slow inactivation of voltage-gated sodium channels I: wild-type skeletal muscle $\text{Na}_v1.4$. *J. Gen. Physiol.* 141, 309–321. <https://doi.org/10.1085/jgp.201210909>.
- Silva, J.R., and Goldstein, S.A. (2013). Voltage-sensor movements describe slow inactivation of voltage-gated sodium channels II: a periodic paralysis mutation in $\text{Na}_v1.4$ (L689I). *J. Gen. Physiol.* 141, 323–334. <https://doi.org/10.1085/jgp.201210910>.
- Silva, J. (2014). Slow inactivation of Na^+ channels. In *Voltage Gated Sodium Channels*, P.C. Ruben, ed. (Springer), pp. 33–49. https://doi.org/10.1007/978-3-642-41588-3_3.
- Qian, K., Yu, N., Tucker, K.R., Levitan, E.S., and Canavier, C.C. (2014). Mathematical analysis of depolarization block mediated by slow inactivation of fast sodium channels in midbrain dopamine neurons. *J. Neurophysiol.* 112, 2779–2790. <https://doi.org/10.1152/jn.00578.2014>.
- Bakkum, D.J., Frey, U., Radivojevic, M., Russell, T.L., Müller, J., Fiscella, M., Takahashi, H., and Hierlemann, A. (2013). Tracking axonal action potential propagation on a high-density microelectrode array across hundreds of sites. *Nat. Commun.* 4, 2181. <https://doi.org/10.1038/ncomms3181>.
- Radivojevic, M., Franke, F., Altermatt, M., Müller, J., Hierlemann, A., and Bakkum, D.J. (2017). Tracking individual action potentials throughout mammalian axonal arbors. *eLife* 6, e30198. <https://doi.org/10.7554/eLife.30198>.
- Bucher, D., and Goaillard, J.M. (2011). Beyond faithful conduction: short-term dynamics, neuromodulation, and long-term regulation of spike propagation in the axon. *Prog. Neurobiol.* 94, 307–346. <https://doi.org/10.1016/j.pneurobio.2011.06.001>.
- Freeman, S.A., Desmazières, A., Fricker, D., Lubetzki, C., and Sol-Foulon, N. (2016). Mechanisms of sodium channel clustering and its influence on axonal impulse conduction. *Cell. Mol. Life Sci.* 73, 723–735. <https://doi.org/10.1007/s00018-015-2081-1>.
- Narahashi, T. (1964). Threshold potential in anodally restored lobster giant axons. *J. Cell. Comp. Physiol.* 64, 239–248. <https://doi.org/10.1002/jcp.1030640208>.
- Adelman, W.J., Jr., and Palti, Y. (1969). The effects of external potassium and long duration voltage conditioning on the amplitude of sodium currents in the giant axon of the squid, *Loligo pealei*. *J. Gen. Physiol.* 54, 589–606. <https://doi.org/10.1085/jgp.54.5.589>.
- Chandler, W.K., and Meves, H. (1970). Evidence for two types of sodium conductance in axons perfused with sodium fluoride solution. *J. Physiol.* 211, 653–678. <https://doi.org/10.1113/jphysiol.1970.sp009298>.
- Chandler, W.K., and Meves, H. (1970). Rate constants associated with changes in sodium conductance in axons perfused with sodium fluoride. *J. Physiol.* 211, 679–705. <https://doi.org/10.1113/jphysiol.1970.sp009299>.
- Chandler, W.K., and Meves, H. (1970). Slow changes in membrane permeability and long-lasting action potentials in axons perfused with fluoride solutions. *J. Physiol.* 211, 707–728. <https://doi.org/10.1113/jphysiol.1970.sp009300>.
- Ruben, P.C., Starkus, J.G., and Rayner, M.D. (1992). Steady-state availability of sodium channels. Interactions between activation and slow inactivation. *Biophys. J.* 61, 941–955. [https://doi.org/10.1016/S0006-3495\(92\)81901-X](https://doi.org/10.1016/S0006-3495(92)81901-X).
- Toib, A., Lyakhov, V., and Marom, S. (1998). Interaction between duration of activity and time course of recovery from slow inactivation in mammalian brain Na^+ channels. *J. Neurosci.* 18, 1893–1903.
- Ellerkmann, R.K., Riazanski, V., Elger, C.E., Urban, B.W., and Beck, H. (2001). Slow recovery from inactivation regulates the availability of voltage-dependent Na^+ channels in hippocampal granule cells, hilar neurons and basket cells. *J. Physiol.* 532, 385–397. <https://doi.org/10.1111/j.1469-7793.2001.0385f.x>.
- Marom, S. (2010). Neural timescales or lack thereof. *Prog. Neurobiol.* 90, 16–28. <https://doi.org/10.1016/j.pneurobio.2009.10.003>.
- Marom, S. (2016). Emergence and maintenance of excitability: kinetics over structure. *Curr. Opin. Neurobiol.* 40, 66–71. <https://doi.org/10.1016/j.conb.2016.06.013>.
- Ori, H., Marder, E., and Marom, S. (2018). Cellular function given parametric variation in the Hodgkin and Huxley model of excitability. *Proc. Natl. Acad. Sci. USA* 115, E8211–E8218. <https://doi.org/10.1073/pnas.1808552115>.
- Gal, A., and Marom, S. (2013). Self-organized criticality in single-neuron excitability. *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* 88, 062717. <https://doi.org/10.1103/PhysRevE.88.062717>.
- Marder, E., and Goaillard, J.M. (2006). Variability, compensation and homeostasis in neuron and network function. *Nat. Rev. Neurosci.* 7, 563–574. <https://doi.org/10.1038/nrn1949>.
- Marder, E., and Rue, M.C.P. (2021). From the neuroscience of individual variability to climate change. *J. Neurosci.* 41, 10213–10221. <https://doi.org/10.1523/JNEUROSCI.1261-21.2021>.
- Zang, Y., and Marder, E. (2023). Neuronal morphology enhances robustness to perturbations of channel densities. *Proc. Natl. Acad. Sci. USA* 120, e2219049120.

30. Liu, Z., Golowasch, J., Marder, E., and Abbott, L.F. (1998). A model neuron with activity-dependent conductances regulated by multiple calcium sensors. *J. Neurosci.* **18**, 2309–2320. <https://doi.org/10.1523/JNEUROSCI.18-07-02309.1998>.
31. O'Leary, T., Williams, A.H., Caplan, J.S., and Marder, E. (2013). Correlations in ion channel expression emerge from homeostatic tuning rules. *Proc. Natl. Acad. Sci. USA* **110**, E2645–E2654. <https://doi.org/10.1073/pnas.1309966110>.
32. O'Leary, T., Williams, A.H., Franci, A., and Marder, E. (2014). Cell types, network homeostasis, and pathological compensation from a biologically plausible ion channel expression model. *Neuron* **82**, 809–821. <https://doi.org/10.1016/j.neuron.2014.04.002>.
33. Migliore, M. (1996). Modeling the attenuation and failure of action potentials in the dendrites of hippocampal neurons. *Biophys. J.* **71**, 2394–2403. [https://doi.org/10.1016/S0006-3495\(96\)79433-X](https://doi.org/10.1016/S0006-3495(96)79433-X).
34. Perge, J.A., Niven, J.E., Mugnaini, E., Balasubramanian, V., and Sterling, P. (2012). Why do axons differ in caliber? *J. Neurosci.* **32**, 626–638. <https://doi.org/10.1523/JNEUROSCI.4254-11.2012>.
35. Zang, Y., and Marder, E. (2021). Interactions among diameter, myelination, and the Na/K pump affect axonal resilience to high-frequency spiking. *Proc. Natl. Acad. Sci. USA* **118**, e2105795118. <https://doi.org/10.1073/pnas.2105795118>.
36. Knowlton, C.J., Ziouziou, T.I., Hammer, N., Roeper, J., and Canavier, C.C. (2021). Inactivation mode of sodium channels defines the different maximal firing rates of conventional versus atypical midbrain dopamine neurons. *PLoS Comput. Biol.* **17**, e1009371. <https://doi.org/10.1371/journal.pcbi.1009371>.
37. Carter, B.C., and Bean, B.P. (2009). Sodium entry during action potentials of mammalian neurons: incomplete inactivation and reduced metabolic efficiency in fast-spiking neurons. *Neuron* **64**, 898–909. <https://doi.org/10.1016/j.neuron.2009.12.011>.
38. Hasenstaub, A., Otte, S., Callaway, E., and Sejnowski, T.J. (2010). Metabolic cost as a unifying principle governing neuronal biophysics. *Proc. Natl. Acad. Sci. USA* **107**, 12329–12334. <https://doi.org/10.1073/pnas.0914886107>.
39. Zheng, Y., Liu, P., Bai, L., Trimmer, J.S., Bean, B.P., and Ginty, D.D. (2019). Deep sequencing of somatosensory neurons reveals molecular determinants of intrinsic physiological properties. *Neuron* **103**, 598.e7–616.e7. <https://doi.org/10.1016/j.neuron.2019.05.039>.
40. Faisal, A.A., White, J.A., and Laughlin, S.B. (2005). Ion-channel noise places limits on the miniaturization of the brain's wiring. *Curr. Biol.* **15**, 1143–1149. <https://doi.org/10.1016/j.cub.2005.05.056>.
41. Navarro, M.A., Salari, A., Lin, J.L., Cowan, L.M., Penington, N.J., Miles, M., and Miles, L.S. (2020). Sodium channels implement a molecular leaky integrator that detects action potentials and regulates neuronal firing. *eLife* **9**, e54940. <https://doi.org/10.7554/eLife.54940>.
42. Carnevale, N.T., and Hines, M.L. (2006). *The NEURON Book* (Cambridge University Press).

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited code		
Code for performing simulations	This paper	Available via ModelDB at https://senselab.med.yale.edu/modeldb/enterCode?model=267610
Software and algorithms		
NEURON 8.1.0	https://www.neuron.yale.edu/neuron/	version 8.1.0; RRID: SCR_005393
MATLAB R2022a	Math Works	RRID: SCR_001622

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and model codes should be directed to and will be fulfilled by the lead contact, Yunliang Zang (yizang@brandeis.edu).

Materials availability

No materials were generated for this study.

Data and code availability

- The model parameter data will be automatically generated when running the simulation code.
- All original code has been deposited at ModelDB and is publicly available as of the date of publication. Simulation code is available at <https://senselab.med.yale.edu/modeldb/enterCode?model=267610>.
- Any additional information required to reanalyze the simulation data reported in this paper is available from the [lead contact](#) upon request.

METHOD DETAILS

In this work, all the simulations were implemented in NEURON⁴² and visualized in MATLAB. The model formulations were the same as in the original Hodgkin-Huxley model and with a new added *i* gate.

$$I_{Na} = \bar{g}_{Na} * g_{Na} * m^3 * h * i * (V - E_{Na}) \quad (\text{Equation 1})$$

$$I_K = \bar{g}_K * g_K * n^4 * (V - E_K) \quad (\text{Equation 2})$$

$$I_{leak} = g_{leak} * (V - E_{leak}) \quad (\text{Equation 3})$$

$$\alpha_m = 0.1 * \left(\frac{(V + 40)}{1 - \exp\left(-\frac{(V + 40)}{10}\right)} \right) \quad (\text{Equation 4})$$

$$\beta_m = 4 * \exp\left(-\frac{(V + 65)}{18}\right) \quad (\text{Equation 5})$$

$$m_{ss} = \frac{\alpha_m}{\alpha_m + \beta_m} \quad (\text{Equation 6})$$

$$\tau_m = \frac{1}{\alpha_m + \beta_m} \quad (\text{Equation 7})$$

$$\frac{dm}{dt} = \frac{(m_{ss} - m)}{\tau_m} \quad (\text{Equation 8})$$

$$\alpha_h = 0.07 * \exp\left(-\frac{(V+65)}{20}\right) \quad (\text{Equation 9})$$

$$\beta_h = \frac{1}{1 + \exp\left(-\frac{(V+35)}{10}\right)} \quad (\text{Equation 10})$$

$$h_{ss} = \frac{\alpha_h}{\alpha_h + \beta_h} \quad (\text{Equation 11})$$

$$\tau_h = \frac{1}{\alpha_h + \beta_h} \quad (\text{Equation 12})$$

$$\frac{dh}{dt} = \frac{(h_{ss} - h)}{\tau_h} \quad (\text{Equation 13})$$

$$\alpha_i = 0.007 * \exp\left(-\frac{(V+65)}{20}\right) \quad (\text{Equation 14})$$

$$\beta_i = 0.1 / (1 + \exp(-(V + 35) / 10)) \quad (\text{Equation 15})$$

$$i_{ss} = \frac{\alpha_i}{\alpha_i + \beta_i} \quad (\text{Equation 16})$$

$$\tau_i = \frac{1}{\alpha_i + \beta_i} \quad (\text{Equation 17})$$

$$\frac{di}{dt} = \frac{(i_{ss} - i)}{\tau_i} \quad (\text{Equation 18})$$

$$\alpha_n = 0.01 * \left(\frac{(V+65)}{1 - \exp\left(-\frac{(V+55)}{10}\right)} \right) \quad (\text{Equation 19})$$

$$\beta_n = 0.125 * \exp\left(-\frac{(V+65)}{80}\right) \quad (\text{Equation 20})$$

$$n_{ss} = \frac{\alpha_n}{\alpha_n + \beta_n} \quad (\text{Equation 21})$$

$$\tau_n = \frac{1}{\alpha_n + \beta_n} \quad (\text{Equation 22})$$

$$\frac{dn}{dt} = \frac{(n_{ss} - n)}{\tau_n} \quad (\text{Equation 23})$$

$g_{Na} = 120 \text{ mS/cm}^2$; $g_K = 36 \text{ mS/cm}^2$; $g_{leak} = 0.3 \text{ mS/cm}^2$; $E_{Na} = 50 \text{ mV}$; $E_K = -77 \text{ mV}$; $E_{leak} = -54.4 \text{ mV}$. $\bar{g}_{Na}$ and $\bar{g}_K$ are scaling factors sampled from Gaussian probability density function. V is in mV and reaction rates are in ms^{-1} . In all models, axial resistivity R_a is $100 \text{ } \Omega\text{cm}$ and C_m is $1 \text{ } \mu\text{F/cm}^2$. We used two different formulations for the slow inactivation gate. In Figure 1, the slow inactivation gate shown above was simulated by scaling the h-gate kinetics by 0.1. In Figures 2, 3, and 4, all the formulations are the same except new i gate formulations are according to Migliore.³³ The impact of slow inactivation is independent of the specific model formulations.

$$i_{ss} = 1 / (1 + \exp((V + 58) / 2)) + i_{min} * (1 - 1 / (1 + \exp((V + 58) / 2))) \quad (\text{Equation 24})$$

$$\tau_i = \exp(0.09 * (V + 60)) / (0.0003 * (1 + \exp(0.45 * (V + 60)))) \quad (\text{Equation 25})$$

$$di / dt = (i_{ss} - i) / \tau_i; \text{ If } \tau_i \leq \tau_{inact}, \tau_i = \tau_{inact} \quad (\text{Equation 26})$$

τ_{inact} is 20 ms except in Figure 4. In Figure 4, i_{ss} was varied by changing i_{min} ; the inactivation time constant at depolarized membrane potentials was varied by changing τ_{inact} ; the recovery time constant at hyperpolarized membrane potentials were varied by a scaling factor, because of its voltage dependence implemented in the model. The axon model has a length of 5 mm. We simulated axonal diameter in the range of $0.1 - 1 \text{ } \mu\text{m}$ (only $1 \text{ } \mu\text{m}$ data were shown), with the segment length of $6 - 120 \text{ } \mu\text{m}$. To avoid the boundary effect, in the beginning part of the model, we added an axon initial segment with the standard Hodgkin-Huxley model channel densities evenly distributed and the stimuli were exerted at the starting point of the axon initial segment. We only analyzed spike propagation in the first 4.5 mm of the axon to avoid potential boundary effect in the distal end.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were conducted using MATLAB and are detailed in the results.